

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GLY 316 — Past Questions & Materials

by Renance

Course: GLY 316

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 2

Levels: 300

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Paper 1 of 2 — GLY 316 Structural Geology

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE

Pas oS,

DEPARTMENT OF GEOLOGY

FIRST SEMESTER EXAMINATION 2024/2025 SESSION.

STRUCTURAL GEOLOGY, GLY 316 CU).

Time Allowed: 2% doars

INSTRUCTION: ANSWER FOUR QUESTIONS IN ALL. Use well labelled and titled diagram(s) to illustrate your answer(s) whicre necessary.

SECTION A

Answer any Two Questions in this Section.

1. a) Define Strain. (7 mark)

(b) Use Mohr diagram to illustrate

(i) Pure shear stress test (ii) 'Triaxial stress test (iii) Uniaxial test. (4% marks)

~~ = (ey Distinguish between thie following pairs; = (42 marksy ~ ~ ~

(i) True dip and Hade

(ii) Antithetic and Synthetic faults

(iii) Myionite and Slickenside

Gv} Horst and Greben

2. (a) Give a genetic classification of faulis and describe stress patterns and orientations assuviaied

2. te

with vhem. (814 marie) j

{b) Mistinguish berween the following pairs: {9 marks)

G) Hemogeneous and heterogeneous strains

(i) Hydrostatic pressuie and hydrostatic tension

Git) Thrust fault and Overthrust fault ;

3. (2) What ic Diastrophism (4 marks)

(b) Siaie the importance of freciures. (6 rarks)

(c) Enumerate two (2) structures that are associated with the following tectonic environments.

i. Convergent setiing. i!. Divergent setting iii. Transform setting. (74 marks)

SECTION &

Ansiver any two Questions in this Section.

1. (a) What is difference between a fold and fold'ng? (3% marks)

(b) Discuss on five parts of a fold, using adiaqram. (5 marks)

ic) Write short notes on six types of based on cynunetry of limb. (9 marks)

2. (a) What is a Foliation? (2% marks)

(0) Discuss three types of spaced foliation. (6 marks)

(c) Write short notes on the following:

(® Recumbent fold. (ii) Closed fold Gi Plunging fold (° mark)

3. (a) What is a lineation? (2% marks)

(v) | What is the difference between parallel, similar and supratenuous folds? (6 marks)

(b) | Write short notes on the following:

(i) Boudins Gi) Crenulation lineation (iii) Slaty cleavage. (9 marks)

Paper 2 of 2 — STRUCTURAL GEOLOGY GLY 316

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE

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DEPARTMENT OF GEOLOGY

FIRST SEMESTER EXAMINATION 2024/2025 SESSION.

STRUCTURAL GEOLOGY, GLY 316 (3CU).

Time Allowed: 2% Hoars

INSTRUCTION: ANSWER FOUR QUESTIONS IN ALL. Use well labelled and titled diagram(s) to illustrate your answer(s) where necessary.

"> SECTION A

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1.(a) Define Strain (mark)

(b) _Use Mohr diagram to illustrate

(i) Pure shear stress test (ii) Triaxial stress test (iii) Uniaxia: test. (4% marks)

(c) Distinguish between the following pairs: ~~ (12 nitrksy

{}) True dip and Hade

(ii) Antithetic and Synthetic faults

(iii) Mylonite and Slickenside

(iv} Horst and Graben

2. (a) Give a genetic classification of faults and describe stress patterns and orientatioas assoviated With jhem. (8% mares) sie

'h) Fistinguish between the following pairs: '9 marks)

(i) Hemogeneous and heterogencous strains

(ii) Hydrostatic pressure and hydrostatic tension

(iii) Thrust fault and Overttirust fault

3.(2} What ic Diastrophism (4-nrarks)

(b) State the importance of fractures. (6 mzarks}

(c) Enumerate two (2) structures that are associated with the following tectonic environments,

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